
Recognizing the predictive boundary of historically restricted physical knowledge

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Abstract

Scientific models have finite domains, normally established only in hindsight, once a successor theory explains the failure. Machine-learning systems for scientific data are optimized to interpolate, to predict forward, or to recover a governing expression; none asks where an established description ceases to hold. Here we show that the reach of a body of physical knowledge can be recovered from that knowledge alone. We restrict a representation to relations recorded before a chosen date, give it no later formula and no example of a departure, and require it to rank measurements that knowledge cannot account for. At an 1899 horizon, an encoder trained on source-audited pre-1900 point clouds ranks twentieth-century physical laws above held-out classical ones with an area under the ROC curve of 0.925, against 0.649 before optimization; an independently frozen extrapolation test reaches 0.888 with no learned representation at all. A discrepancy found this way is tied to the incumbent model rather than to the data: on historical measurements it collapses when, and only when, the successful successor prediction is admitted. The same test can be moved to any period for which a dated corpus exists.

1 Introduction

Physics advances by retiring descriptions that have stopped working. The Rayleigh–Jeans law, the Dulong–Petit law and the classical account of electrical resistance were each accurate across the range in which they were built and each failed outside it, and in every case the failure was recognized only once a successor theory had supplied the explanation. The domain over which a model can be trusted is therefore something the field has established in retrospect.

Machine-learning systems for scientific data are built for other targets. They are optimized to interpolate within a sampled range, to predict forward, or to recover a compact governing expression from observations [1, 2]. Conformal prediction calibrates residuals, change-point methods locate distributional shifts, and Bayesian model discrepancy compares observations with a simulator [3–5]. Symbolic regression searches for a replacement expression once a discrepancy has already been found [6, 7].

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None of these operations specifies which body of physical knowledge is under test. A residual is large relative to some fitted model, a change point is a shift in some distribution, and a recovered expression is a fit to some data; in each case the knowledge being audited is implicit in the setup rather than declared. Nor can shape novelty close the gap, because a physically decisive departure may keep an entirely ordinary functional form while a familiar mechanism may produce a highly unusual one.

Here we make the knowledge horizon an explicit input and test it directly with EPOCH (Epistemic Probe of Classical Horizons). Every test declares its admissible model set, fitted parameters, observation and censoring model, fitting data, complete adaptive procedure, statistic and error rate, and returns one of four verdicts: compatibility over the measured range, a localized boundary with an uncertainty interval, incompatibility over the sampled range with no identifiable boundary, or abstention. Three kinds of evidence are reported separately rather than averaged into a single score: a positive-only representation of the declared corpus, data-anchored extrapolation from a sampled incumbent regime, and a theory-conditioned residual test against a case-matched null.

We set the horizon at 1899, the eve of the largest replacement of a model set in the history of physics and the last date at which the classical corpus can still be reconstructed from dated primary sources. Every admitted training relation is linked to a dated source-law family, splits are grouped by family, and the learned optimization contains neither later formulas nor artificial breakdown templates; the large-scale protocol was timestamp-frozen before any temporal score was computed. We find that a representation built this way ranks twentieth-century formula families above held-out pre-1900 groups, that explicit extrapolation reaches the same conclusion without a learned representation, that the discrepancy found on five historical datasets collapses only under the successful successor prediction and not under wrong-shape or wrong-mechanism alternatives, and that on real measured series the leading source of variation is the observation process rather than the period of the underlying physics.

2 Results

2.1 Declaring the object of a boundary test

EPOCH returns one of four verdicts for a set of measurements and a declared model set: compatible over the measured range; localized model incompatibility with an uncertainty interval; model incompatibility over the sampled range with no identifiable boundary; or insufficient evidence (Fig. 1). Which evidence produces the verdict is fixed in advance. Data-anchored evidence is used when a clean incumbent interior is observed, theory-conditioned evidence otherwise if an incumbent prediction and an uncertainty model are available, and the test abstains when neither route is supported.

The three components are calibrated against their own reference distributions and reported separately rather than averaged. The data-anchored component receives a notation-free point cloud, fits the best admissible generic form on a contiguous interior regime and tests its extrapolation at the frontier through split-conformal nonconformity and an extrapolation-residual ratio over power, exponential, linear and constant fits. The theory-conditioned component forms measurement-model-aware residuals against a named incumbent prediction and tests them against a case-matched counterfactual null. The learned component measures distance from a positive-only representation of relations recorded before the declared date. Every test declares its admissible model set, fitted parameters, observation and censoring model, fitting data, complete adaptive procedure, discrepancy statistic and error rate, and a named physical-theory conclusion follows only when the model set and observation model encode that theory’s relevant predictions (Methods). Table 1 collects the principal quantitative results reported below.

2.2 Recognizing later physics from pre-1900 relations alone

To ask how much of a knowledge boundary is visible in the shape of a relation alone, we trained a 20.9-million-parameter encoder on 880,896 normal-law point clouds derived from

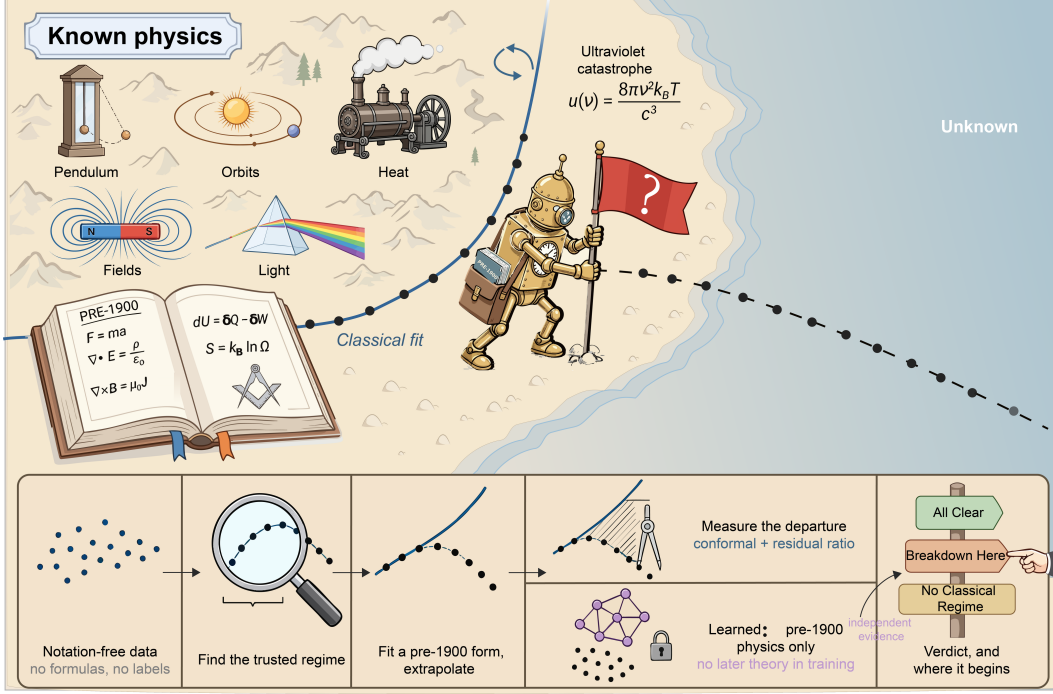


Figure 1. Testing a declared knowledge horizon. Top, a system restricted to physics recorded before 1900 carries that corpus to the edge of what it predicts. An admitted classical form fitted inside the sampled regime (blue) tracks the measurements until they leave it; beyond that point the extrapolation and the data separate, and the question is whether the separation can be measured and located. Bottom, the test as applied here. A notation-free point cloud carrying no formula, unit or label enters at the left. A fixed rule identifies a contiguous interior regime and fits an admitted pre-1900 form there, the extrapolation is compared with the frontier through split-conformal nonconformity and an extrapolation-residual ratio, and a positive-only encoder trained on pre-1900 relations alone supplies an independent representation distance. The signpost carries the verdict and, where an incumbent regime is sampled, the location at which compatibility is lost; a fourth output, abstention, applies when neither evidence route is supported.

3,441 source-audited pre-1900 formula reductions. The model aligns a permutation-invariant point-cloud student to a training-only formula-graph teacher; at inference it receives an unordered normalized point cloud and nothing else, with no formula, graph, units, date, source or anomaly label. Data views, checkpoint rule, score construction, aggregation and bootstrap procedure were timestamp-frozen before any temporal score was computed.

Against nine held-out pre-1900 historical-family groups, the 1899-horizon score separates 38 citation-anchored families dated 1901–1950 with AUROC 0.921 [0.833, 0.994], 17 post-1950 families with 0.935 [0.823, 1.000] and their 55-family union with 0.925 [0.851, 0.982] (Figs. 2a and 3). The same-seed, architecture-matched model before optimization reaches 0.589, 0.784 and 0.649. All four trained score components separate the 55-family union on their own, with AUROCs from 0.800 to 0.925.

Sampling alone could in principle produce that separation, because the later families were generated by a different process from the internal pre-1900 corpus. Regenerating all nine internal groups under the later benchmark’s point count, x-jitter, repeat count, noise range and heteroscedastic profile leaves AUROC at 0.885 [0.778, 0.965] for the 38 near-future families and 0.857 [0.741, 0.945] for all 55, against 0.516 and 0.545 before optimization. Pre-1900 optimization therefore contributes beyond the generator contrast, with the nine negative families governing the interval.

The gain is specific to the historical corpus rather than to function exposure in general. A source-audited reconstruction supplies 268 point clouds from 51 cited pre-1900 law families, split by family into 205/17/46 training, validation and test records, and a compact encoder

Table 1. Principal quantitative results.

Analysis	Result
Large learned model, 38 families from 1901–1950 vs nine pre-1900 groups	AUROC 0.921 [0.833, 0.994]; unoptimized 0.589
Large learned model, 55 families from 1901–1996 vs nine pre-1900 groups	AUROC 0.925 [0.851, 0.982]; unoptimized 0.649
Generator-matched learned analysis, 55 later vs nine pre-1900 groups	AUROC 0.857 [0.741, 0.945]; unoptimized 0.545
Independently frozen four-form analysis, 55 later vs 11 pre-1900 families	AUROC 0.888 [0.800, 0.959]
Source-audited compact encoder vs matched generic-function encoder	Family AUROC 0.903 [0.729, 1.000] vs 0.375 [0.167, 0.618]
Blinded transfer, 200 classical records and 30 synthetic departures	AUROC 0.88
Calibrated data-anchored decision, 12 departure and 12 classical families	TPR 0.271 [0.073, 0.500]; 0/96 false positives at the 5%-targeted threshold
Clock advancement on identical data grids	4/5 old-horizon mismatches clear under the successful successor prediction
Horizon-specificity controls	5/5 wrong-shape and 5/5 wrong-mechanism predictions remain rejected
Frozen components on sixteen measured series, 5 post-1900 vs 9 classical or metrological groups	Learned AUROC 0.62 [0.29, 0.89]; four-form AUROC 0.78 [0.38, 1.00]
Boundary localization on 900 held-out simulations	90% interval coverage 91.3% [89.3%, 93.0%]
Observation-process stress test at maximum severity	FPR 0.40 for saturation; 0.63 for censoring; ≤ 0.11 for all other tested effects

104 trained on the 205 classical records without deformation negatives reaches family AUROC
105 0.903 [0.729, 1.000], against 0.329 and 0.288 for matched random features under kNN and
106 shrinkage Mahalanobis. A capacity- and data-matched encoder trained instead on generic
107 smooth expressions, with identical architecture, seed, schedule, record count, family-size
108 profile and functional-signature histogram, reaches 0.375 [0.167, 0.618]; the paired difference
109 is 0.528 [0.264, 0.764], and across three paired seeds the two ranges are 0.889–0.910 and
110 0.292–0.403. A unit-preserving deformation audit gives source-family-balanced AUROC
111 0.684 with dimensional labels and 0.922 without them, so the zero-unit ranking is not carried
112 by units.

113 2.3 Explicit extrapolation as independent evidence

114 Explicit extrapolation reaches the same conclusion without a learned representation. A
115 citation-anchored audit contains 55 named physical-law families first valid from 1901 to
116 1996, with eight independently sampled and noisy 120-point realizations each, giving 440
117 formula-generated point clouds (Fig. 2b); eleven cited pre-1900 families excluded from
118 encoder training serve as controls. Neither component is retrained or recalibrated here, the
119 zero-unit encoder, its pre-1900 reference bank and the four-form statistical scorer being
120 frozen beforehand, and all quantities are reduced to one median per source-law family.

121 The four-form component ranks later families above the earlier holdouts with AUROC 0.888
122 [0.800, 0.959], and the compact learned component with 0.669 [0.534, 0.798]. A blinded
123 evaluation over 200 historical formula records and 30 synthetic realizations of five breakdown
124 families gives 0.88, with recall 0.90 and a false-positive rate of 0.24 at the 0.5 screening
125 threshold.

126 Functional structure explains part of the ranking. Learned and statistical AUROCs are
127 0.629/0.795 for 12 later affine, power or exponential laws, 0.600/0.982 for five threshold
128 or piecewise families, and 0.909/0.864 for six nonmonotone or oscillatory families, so the
129 two scores respond differently to smooth familiar forms, to thresholds and to nonmonotone
130 curves.

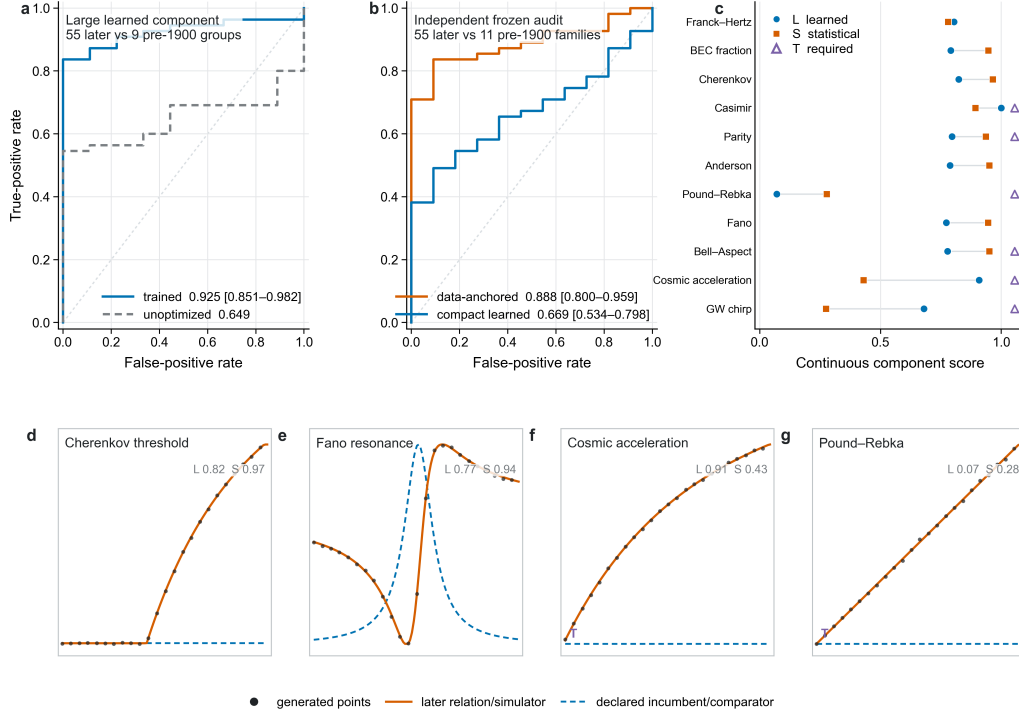


Figure 2. Temporal transfer and component scores across later physics. **a**, ROC curves for the 20.9-million-parameter encoder trained on pre-1900 point clouds (blue) and the same architecture before optimization (grey), comparing 55 later formula families with nine pre-1900 groups. **b**, Independent audit of the data-anchored component (orange) and compact learned encoder (blue) against 11 cited pre-1900 families. **c**, Learned scores (L, circles) and size-matched statistical percentiles (S, squares) for 11 development cases. Triangles mark cases routed to an incumbent-conditioned test (T). **d–g**, Point clouds, later relations and incumbent comparators for Cherenkov emission [8], Fano resonance [9], cosmic acceleration [10] and Pound–Rebka redshift [11]. Checkpoints and calibration objects remain fixed throughout.

Applied without retraining to 11 further twentieth-century cases (Fig. 2c–g), the two coordinates diverge again: Casimir plates [12] give 1.000/0.892, Cherenkov emission [8] 0.823/0.965, cosmic acceleration [10, 13] 0.908/0.429 and Pound–Rebka redshift [11] 0.071/0.277. Scalar or sparse-level claims, including the Lamb shift [14] and the electron anomalous magnetic moment [15], route to the theory-conditioned component instead, and the primary AUROC is unchanged (Supplementary Note 17).

2.4 Historical measurements at the 1899 horizon

Four historical datasets cover the two outputs available when the incumbent is a named classical law: whole-range model rejection and boundary localization (Fig. 4a–d). The initial data-anchored screen gives percentiles of 0.95 for FIRAS, 0.95 for copper specific heat, 0.66 for Bertozzi and 0.66 for Onnes; case-specific tests then determine which output the measurements support. Supplementary Note 5 gives finite-sample tail probabilities and the multiplicity calculation.

In the 43-point COBE-FIRAS cosmic-microwave-background monopole spectrum [16], a Rayleigh–Jeans law [17, 18] proportional to ν^2 over-predicts measured intensity above the peak by 475-fold to 3,000-fold across fitting windows; the 20% window gives a bootstrap interval of [1,500, 2,400]. At $T = 2.725$ K, FIRAS samples $x = h\nu/kT$ from 1.2 to 11.3 rather than the pure Rayleigh–Jeans regime $x \ll 1$, so the output is whole-range incompatibility with the declared classical prediction rather than a located boundary.

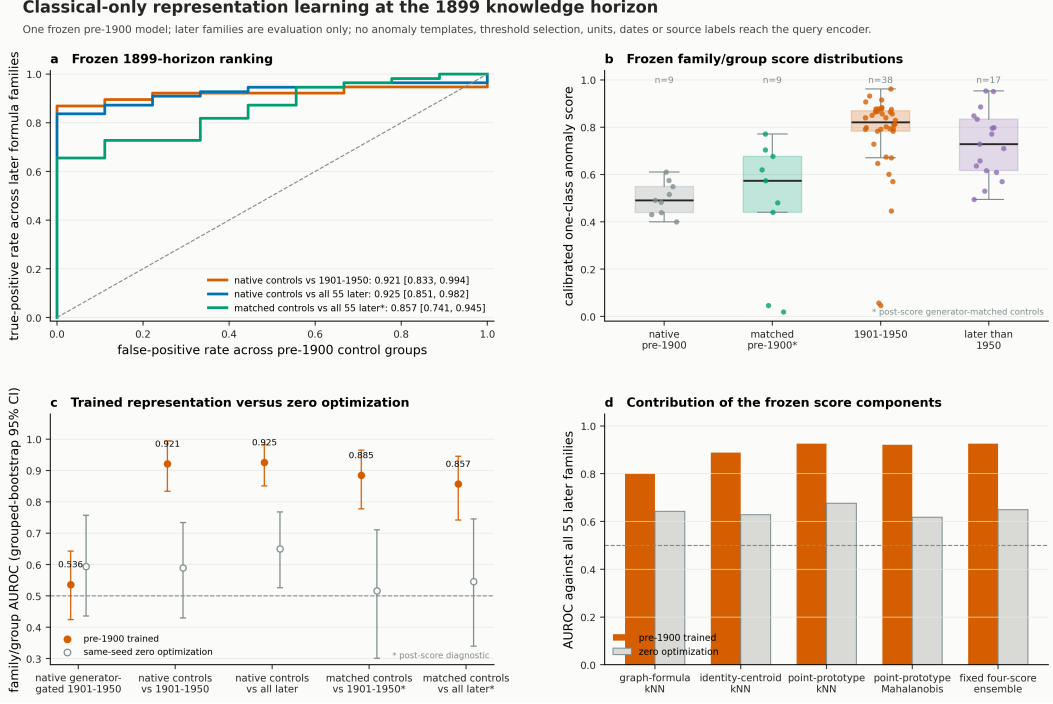


Figure 3. Large-scale representation audit at the 1899 horizon. **a**, The frozen model separates 38 families dated 1901–1950 and all 55 later families from nine pre-1900 groups. All-later AUROC is 0.857 [0.741, 0.945] against generator-matched controls. **b**, One-class score distributions for native and matched pre-1900 controls and the two later-era strata. **c**, AUROCs for trained and same-seed zero-optimization models. The native generator-gated comparison is shown separately. **d**, Each of the four distance components and their fixed equal-weight mean improves over zero optimization on the 55-family comparison. Inferential units are source-law families or frozen relation groups.

Bertozzi’s electron measurements[19] report kinetic energy against $\beta^2 = v^2/c^2$. The classical law $\frac{1}{2}mv^2$ crosses $\beta^2 = 1$ near 0.26 MeV and reaches $\beta^2 = 58.7$ at 15 MeV, equivalent to $7.7c$, whereas the measurements approach $\beta^2 = 1$. The lowest point is already $0.87c$, placing the complete series outside a non-relativistic interior, and EPOCH rejects the Newtonian energy–velocity prediction over the sampled range.

The Dulong–Petit law [20] predicts temperature-independent molar heat capacity, whereas the NIST cryogenic reference curves for six solids fall at low temperature. Five materials contain a high-temperature plateau and support candidate localization relative to the constant model; beryllium supports whole-range incompatibility. Interior R^2 ranges from 0.965 to 0.998 and departure ratios from 127 to 1,490, and for OFHC copper the classical extrapolation over-predicts the 4 K reference value by roughly 3.7×10^3 . The reference-polynomial curves share one low-temperature mechanism and are grouped accordingly.

Onnes’s mercury resistance values [21], digitized from $R(T)$ and anchored by the reported resistance below $10^{-5} \Omega$, show a smooth normal-state trend followed by a discontinuous drop. Relative to the declared smooth and residual-resistance models [22], EPOCH localizes a candidate boundary near 4.20 K, with figure-reading uncertainty and censored lower values carried into the case-specific evidential status.

Millikan’s 1916 photoelectric measurement shows why shape evidence and theory evidence must be reported apart [23]. Stopping potential is linear in frequency, $V_{\text{stop}} = (h/e)(\nu - \nu_0)$, with $R^2 = 0.996$ and a slope that recovers h/e to 0.3% (Fig. 4e), so the generic dictionary accepts the shape without objection. The same four measurements reject the classical prediction of frequency-independent photoelectron energy. Their fitted intercept gives

$\nu_0 = 4.39 \times 10^{14}$ Hz, and because all four frequencies lie above ν_0 the output is whole-range model rejection rather than localization of the emission threshold.

Three measured controls span two branches of physics and three and a half centuries of measurement (Supplementary Fig. 1a–c). Independently measured Solar-System orbits recover Kepler’s exponent of $3/2$ [24] to four figures (1.500) and score 0.02, the four Galilean moons give exponent 1.500 and score 0.00, and Boyle’s original 1662 pressure–volume table [25] gives exponent -0.998 and scores 0.38. All three fall below the 0.5 screening threshold.

2.5 Advancing the horizon on the same measurements

A calibrated mismatch is informative only if it tracks the declared model rather than the data, so each dataset is scored twice on the same grid and under the same error model: first under its old incumbent prediction, then under the historically successful successor. The old horizon is rejected in all five cases (bootstrap $p = 0.001$; Fig. 5a). The advanced horizon clears FIRAS under Planck’s law [26] ($p = 1.000$), Bertozzi under relativistic kinetic energy [27] ($p = 0.089$), Onnes under a superconducting transition model ($p = 1.000$) and Millikan under the photoelectric relation [28] ($p = 0.900$), with mismatch statistics falling by factors from approximately 2.0×10^4 to 7.3×10^{10} .

Copper heat capacity is the exception and is sensitive to the error model. A Debye-plus-electronic form [29] reduces mismatch by 3.86×10^3 but remains rejected under a 5% reference-curve error floor ($p = 0.001$), clearing at 7.5%. Across all five cases, low-dimensional wrong-shape families remain rejected after Benjamini–Hochberg correction ($q \leq 0.0015$), as do dimensionally coherent but mechanism-irrelevant physical forms (1,000 bootstrap replicates; $p = q = 1/1001$). Four of the five score collapses are therefore specific to the successful successor prediction rather than to any better-fitting curve.

2.6 Measured series at the same horizon

Frozen components applied to sixteen unmodified NIST measurement series rank the five post-1900 mechanism groups above the nine classical and metrological groups with AUROC 0.62 [0.29, 0.89] for the learned score and 0.78 [0.38, 1.00] for the four-form score (Fig. 6 and Supplementary Note 18). For each series the column semantics, the 1899 incumbent and the mechanism group that serves as the inferential unit are declared from its primary source before scoring; neither component is retrained or recalibrated, measured series pass through a rescale-only view function that adds no synthetic noise, and all 128 measurement-view scores are released.

Copper thermal expansion measured from 24 K to 852 K takes the highest post-1900 learned score, its coefficient falling towards zero below the Debye temperature in a way classical equipartition of lattice modes cannot produce [30], and the density profile of a cold Bose gas [31, 32] takes the highest four-form score. A sulfur Rydberg series [33], whose inverse-square form was available empirically in 1888, belongs to the classical class yet ranks above four of the five post-1900 groups.

Two series bracket what the components read from measured data. Circular-interference transmittance rises to a sharp resonance at 451 nm, the most unusual shape in the collection, yet scores low on both components, so morphological novelty by itself moves neither coordinate. Superconducting magnetization relaxation, which no pre-1900 theory accounts for, is close to logarithmic in time and takes the lowest four-form score of any series. Across the whole collection the two highest learned scores belong to a five-year standard-resistor record and a 23-point polymer-impedance curve: on measured data the leading axis of variation is the observation process rather than the period of the underlying physics. Fourteen mechanism groups set the width of both intervals.

2.7 Calibration, thresholds and the observation process

On the 36-dataset fair suite the statistical detector reaches AUROC 0.80 [0.64, 0.94], against 0.58 for Gaussian-process discrepancy [34] and 0.77 for split-conformal prediction

(Supplementary Fig. 1d,f). Withholding the classical law, or both the law and the interior regime, leaves AUROC at 1.00 and 0.97, so the result does not depend on supplying the answer.

Calibrating the data-anchored score separately at $n = 5, 9, 43$ and 120 over 12 departure and 12 classical families fixes the operating point. At $n = 120$, TPR is 0.271 [0.073, 0.500] at the frozen 5% quantile and 0 at the 1% quantile, with a realized false-positive rate of 0/96 at both. Sensitivity is 1.0 for smooth roll-off and discontinuous-drop simulations and zero for seven families, so this operating point buys specificity at a known cost in power.

Altering the observation process while holding the underlying classical relation fixed shows where that specificity fails. At the strongest simulated levels a detector calibrated to 5% reports false-positive rates of 0% for calibration drift, 40% for sensor saturation, 63% for censoring, 9% for x-axis error, 11% for outliers, 2.5% for correlated noise and 0% for heteroscedasticity (Fig. 5b). The output schema therefore carries a measurement-process branch, with saturation and censoring tested explicitly for plateau and threshold cases.

Boundary estimates need the same treatment. Ordinary measurement bootstrap intervals cover only 53.9% of true onsets at a nominal 90% level, and 8.5% for steps; size-conditional split-conformal radii calibrated on 899 independent simulations raise held-out coverage to 91.3% [89.3%, 93.0%] across 900 simulations, with family-wise coverage from 87.8% for smooth turnovers to 94.4% for kinks (Fig. 5c). The reported boundary is therefore an interval rather than a single estimated onset.

Simulations extend the evaluation into incumbent regimes absent from the FIRAS and Bertozzi measurements. Blackbody radiation spanning $x = h\nu/kT$ from well below to well above unity contains a Rayleigh–Jeans interior with $R^2 = 0.99$ followed by a departure with conformal score 468 and ratio 192; relativistic kinetic energy yields a localized boundary at 0.86c, and simulated photoelectric voltage localizes the frequency threshold.

3 Discussion

We introduced a test of a declared knowledge horizon and applied it at 1899. Trained only on source-audited pre-1900 relations, with no later formula and no breakdown template in its optimization, a 20.9-million-parameter encoder ranks 55 twentieth-century formula families above nine held-out pre-1900 groups at AUROC 0.925, against 0.649 for the same architecture before optimization; regenerating those controls under the later benchmark’s sampling and noise process leaves 0.857, and an independently frozen four-form scorer reaches 0.888 on the same comparison.

A mismatch is interpretable only when it is tied to a named incumbent model. Scoring each historical dataset twice on the same grid and under the same error model, four of five old-horizon discrepancies disappear once the successful successor prediction is admitted, while wrong-shape and dimensionally coherent wrong-mechanism alternatives stay rejected after multiplicity correction. Copper heat capacity is the exception, and its behaviour is informative: the advanced model clears only when the reference-curve error floor is relaxed from 5% to 7.5%, so the observation model belongs to the knowledge horizon rather than to the nuisance parameters. For a closed-form theory, advancing the horizon amounts to replacing the admitted prediction and recalibrating its reference distribution, and simulator-based references give the corresponding construction where no closed form exists.

Shape novelty and predictive failure are not the same quantity. Later affine, power and exponential laws often stay close to the generic dictionary, whereas threshold, piecewise, nonmonotone and oscillatory relations separate strongly; conversely a physically decisive discrepancy can keep an entirely ordinary shape, as Millikan’s stopping potential and the Pound–Rebka redshift both do. The measured series make the point from both directions. A classical circular-interference resonance, the most unusual morphology in that collection, scores low on both components, while superconducting flux creep [35], which no pre-1900 theory accounts for, is close to logarithmic in time and sits at the bottom of the four-form ranking. Keeping the component scores separate rather than averaging them is what preserves this distinction.

Several limitations bound what these results establish. Although the 1899 experiment is internally controlled, its negative class is small: nine held-out pre-1900 groups carry the primary learned comparison and set the width of its interval, and the generator-matched analysis was registered after the primary scores were opened. The native generator-gated comparison gives AUROC 0.536, so the sampling process remains an active component of the native contrast and the matched value of 0.857 is the estimate that controls for it. On measured series the ranking falls to 0.62 for the learned component and 0.78 for the four-form component, with intervals that span chance, and the two highest learned scores in that collection belong to a standard-resistor record and a polymer-impedance curve rather than to any physical departure — the same sensitivity that saturation and censoring expose in the simulated instrument tests.

EPOCH draws on neural extrapolation [1, 2], split-conformal prediction, Bayesian model discrepancy and change-point detection [3–5], and shares machinery with model-independent searches in particle physics[36] and with machine rediscovery of natural laws [6, 7, 37]. What differs is the target of inference: EPOCH tests the predictive domain of a declared model before asking symbolic regression or theory construction for a replacement. Once a calibrated mismatch exists, that search can proceed under a parsimony prior [38], as the Bayesian machine scientist and AI Feynman do in recovering Planck-type expressions from blackbody-type data [6, 7], and physical interpretation then supplies the mechanism behind the fitted form.

Large libraries of measured relations can be ranked against a frozen horizon, and selected cases rerun with explicit measurement errors and incumbent predictions to obtain a model test and, where an incumbent regime is sampled, a boundary interval. Moving the interface to another period requires only a different dated corpus, admitted prediction and calibration set. The decisive next experiment freezes the full decision path in advance and runs it against mechanism-independent measurements whose instrument models have been audited, so that the observation process is known rather than assumed.

4 Methods

4.1 Declaring a test

An EPOCH test is the tuple $\mathcal{T} = (\mathcal{M}, \Theta, P_\varepsilon, I, A, S, \alpha)$, in which $\mathcal{M}$ is the admissible model set, Θ the fitted parameters, P_ε the observation and censoring model, I the fitting data, A the complete adaptive procedure, S the discrepancy statistic and α the error rate. Because direction choice, window search, model selection and parameter fitting all belong to A , each of them is rerun inside every calibration and null replicate rather than being fixed once on the test data. Confirmatory tests use the finite-sample upper-tail probability $p = (1 + \#\{S_i \geq S_*\})/(n_{\text{cal}} + 1)$ with ties counted conservatively, and dataset scans report Benjamini–Hochberg adjusted q -values [39]. The primary endpoint is the true-positive rate at a threshold targeted to 5% false positives on independent calibration, with the 1% threshold secondary; the 0.5 percentile used in the historical screen is a screening threshold and not a rejection criterion. Exchangeability is defined at the level of the source-law family or mechanism family, so the guarantee that follows applies to the declared exchangeable generator rather than to an arbitrary frontier.

4.2 Generic dictionary and the interior–frontier split

The data-anchored component draws on a fixed parsimonious vocabulary of power, exponential, linear and constant forms. An eight-form ablation adding quadratic, cubic, square-root and logarithmic terms was tested and discarded, because the four-form version extrapolates better on validation data. Given a point cloud, a fixed rule takes the candidate interior to be the contiguous regime in which one admitted form fits with high internal R^2 , and the frontier to be the complementary region; the search runs from both ends and fit quality decides whether the low- or the high-value direction is retained. Split-conformal nonconformity [40] is then computed from interior residual quantiles, and the extrapolation-residual ratio compares the median log-residual at the frontier with its interior value. Each form is fitted from several initializations, and both denominators carry a one-percent relative floor so that a near-perfect

interior fit cannot produce an unbounded ratio. A capped variant, which averages rescaled components clipped at 3, is retained for comparison with earlier results. Relations with fewer than 20 points are calibrated against size-matched simulations.

4.3 Large-scale pre-1900 representation

The scaling experiment uses a 20,908,547-parameter dual encoder trained without any anomaly supervision. Its query branch is a permutation-invariant six-block Set Transformer[41] of width 384 with eight heads, four pooling tokens and a 256-dimensional normalized output, and during positive-only pretraining it is aligned to a formula-structure teacher built from four edge-aware graph updates and two graph-local attention layers, one of whose update networks is a 96-channel eight-basis RBF-KAN module [42]. At query time the encoder receives a median/IQR-normalized unordered (x, y) cloud and nothing else: no equation, graph, units, date, law name, post-cutoff label or deformation example.

Training draws on 880,896 normal-law clouds derived from 3,441 audited pre-1900 formula reductions, grouped into 91 contrastive identities that span 64 historical-family tags, with held-out calibration and internal sets of 11,648 and 62,336 clouds from 91 and 487 formulas. Reducing these to scientific groups leaves nine historical-family groups, which form the internal control class for every AUROC reported here. The model trains for 30 epochs at batch size 1,024 under a checkpoint rule fixed before any temporal evaluation. Its one-class score averages four calibrated ranks — top-eight cosine distance to graph-formula embeddings, top-eight distance to identity centroids, top-eight distance to point-formula prototypes, and shrinkage-Mahalanobis distance [43, 44] to those prototypes — each transformed by its own pre-1900 calibration empirical CDF. Clouds are reduced by median to formulas and then to source-law families or frozen relation groups, and family- and group-level AUROCs use 2,000 grouped-bootstrap replicates. An architecture-matched, same-seed checkpoint that has received no optimization passes through the identical pipeline. Code, configuration, score construction, seed and checkpoint rule were timestamp-frozen before the first temporal score was computed.

For the generator-matched analysis, all nine internal control families were regenerated with the temporal benchmark’s point count, x-jitter, repeat count, noise range and heteroscedastic profile, while the checkpoint, reference memories, calibration CDF, score and aggregation were left unchanged; the same analysis was then repeated with the zero-optimization checkpoint. A separate 96,064-cloud generator-gated set and the cumulative pre-1950 pilot are reported in Supplementary Note 16.

4.4 Compact encoder and its controls

The compact set transformer has width 128 and three induced-set-attention blocks, and trains for 1,000 steps on two independently subsampled and rescaled views of 205 source-audited classical point clouds drawn from 40 families. Families are sampled without replacement so that repeated laws cannot become false contrastive negatives, and an auxiliary head predicts functional signatures. No breakdown, saturation, step, kink, splice or crossover example enters this optimization. Matched models are trained with SI dimensions and with zeroed dimension metadata, and the zero-unit model is primary for the synthetic benchmark. Frozen one-class kNN and shrinkage-Mahalanobis scores are compared against a matched random encoder. Cross-conformal calibration removes each training family in full, reduces the eight measurement views within a record and then reduces records to one median per family; the higher-order 95th-percentile statistic over the 40 leave-family-out family scores is frozen before evaluation, and hash-linked training and evaluation files enforce checkpoint and data identity.

Three controls bound what this representation has learned. A second zero-unit encoder shares architecture, seed, batch size, augmentation, auxiliary objective and 1,000-step schedule but is trained on 205 point clouds from globally valid smooth expressions in 40 pseudo-families, matched to the historical set in record count, family-size profile and the histogram of eight-bit functional signatures. Both encoders are retrained from scratch under three paired initialization and augmentation seeds while the evaluation-view seed and the 24 mechanism families stay fixed, and we report the observed range and sample standard deviation across

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those runs without altering the canonical checkpoints. Finally, each of 46 held-out point clouds is paired with saturation, step, kink, splice, roll-off and crossover deformations that retain its archived input and output SI vectors; scores reduce to one clean and six deformation values within each of seven source-law families, the AUROC difference is bootstrapped by family, and each checkpoint is read against its own higher-order 95th-percentile threshold. Thirty-four of those records carry at least one nonzero dimensional exponent.

Routing for the learned component is fixed in advance. Point clouds with at least 20 observations receive a learned score, near-constant responses route instead to a theory-conditioned constant-or-noise model, and formula and deformation families are the resampling units throughout. Where a deformation-supervised probe is quoted for comparison, it carries explicit anomaly-shape priors, whereas the positive-only encoders reported as primary see normal pre-1900 relations alone.

4.5 Frozen post-1900 formula transfer

The transfer registry contains 55 source-law families first valid between 1901 and 1996, each carrying a primary historical citation and 45 of them a DOI, spanning thermionic emission [45] through discrete atomic excitation [46], parity violation [47], absence of diffusion in disordered lattices [48], the local-hidden-variable bound and its experimental test [49, 50] and the first gravitational-wave detection [51]. Every law is represented by a disclosed one-dimensional nondimensional reduction over a fixed support, realized eight times at 120 points with varying nuisance parameters, sampling and 0.5–2.5% noise, giving 440 records across 55 inferential units. Special reductions, among them the BCS interpolation [52], the two-flavour neutrino survival curve [53] and the ideal quantum-Hall staircase [54], are marked as such in the registry, and offline checks cover family and seed uniqueness, citation-year consistency, finite nonconstant curves, point ordering, file hashes and separation from the pre-1900 archive.

No training or calibration takes place during transfer. The checkpoint and the pre-1900 data hash must match those of the cross-family audit before scoring begins. The learned score is reduced by median over measurement views, records and families and read against the pre-existing higher-order q_{95} statistic from the 40 leave-family-out calibration families, while the statistical score reruns the full four-form selection and extrapolation procedure against the existing $n = 120$ thresholds. Eleven cited pre-1900 families serve as controls, AUROC intervals resample families, and twelve of the post-1900 laws deliberately overlap the declared affine, power or exponential forms. Per-family conformal p -values and component union and intersection rates are reported as secondary diagnostics.

4.6 Historical measurements and horizon-specificity controls

We analyse the published COBE-FIRAS monopole spectrum with its pointwise errors, five Bertozzi values digitized from the 1964 report, NIST specific-heat reference polynomials for six materials, Onnes resistance values including upper limits, Millikan sodium stopping potentials and Solar-System orbital constants. Each case is serialized as a test object recording input status, model set, fitted parameters, observation model, fit subset, adaptive procedure, decision rule, output and provenance. Old- and advanced-horizon counterfactual nulls are evaluated on the same x -grid, and explicit error floors stand in for unavailable covariance and for digitization uncertainty and are varied in sensitivity analysis.

To establish that a collapse in mismatch is specific to the successful successor, wrong-shape and dimensionally coherent wrong-mechanism forms are refitted on the same case grids and observation models with 1,000 parametric-bootstrap replicates. The physical controls substitute fermionic occupation for photon statistics, a proton for the electron rest-energy scale, an Einstein oscillator [30] for Debye acoustic modes [29], Arrhenius transport for the superconducting transition, and Compton recoil for the photoelectric slope; dimensional ratios and parameter units are retained in the machine-readable output.

4.7 Measured series

Sixteen series are taken unmodified from the NIST Dataplot and Statistical Reference Dataset collections and bound to their SHA-256 hashes. A registry declares, for each series, the numeric block, the abscissa and response columns, the 1899 incumbent, the mechanism group and the horizon class before any score is computed. Where a file header and its stored block disagree on column order, the order is resolved against certified values, and a file that is a subset or decimation of another series shares that series' mechanism group rather than forming a second unit. Series are sorted by abscissa and thinned uniformly to at most 2,000 points. The learned component applies the declared near-constant routing rule and a rescale-only view function that adds no synthetic noise to a measurement, with the reference bank rebuilt from the frozen pre-1900 training split; the statistical component reruns the complete four-form direction, window, model-selection and scoring procedure on eight subsamples drawn at the largest frozen calibration size each series supports. Scores reduce by median within a series and then within a mechanism group, and the grouped AUROC resamples mechanism groups.

4.8 Synthetic suites, baselines and localization

The held-out evaluation comprises 180 labelled datasets and the fair baseline suite 36, on which split-conformal prediction, the extrapolation-residual ratio, Gaussian-process discrepancy [34], CUSUM [4] and RESET [55] are compared. The mechanism-level benchmark contains 12 classical and 12 breakdown families with eight realizations each; its 5% and 1% thresholds come from 785 valid size-matched calibration simulations, and uncertainty is resampled by mechanism family rather than treating noise realizations as independent phenomena. The learned checkpoint audit uses 192 datasets over the same 24 families. Instrument tests apply calibration drift, sensor saturation, censoring, x-axis error, outliers, correlated noise and heteroscedasticity to classical synthetic relations while leaving the underlying law intact, and boundary intervals are obtained by split-conformal calibration of the normalized localization error separately at $n = 40, 80$ and 160 .

4.9 Real-measurement inventory

Forty-four files were downloaded from the NIST Dataplot and Statistical Reference Dataset collections and their SHA-256 hashes frozen. The inventory holds 20 instrument controls, 11 classical controls, six modern-known controls and seven horizon cases. A dataset is promoted into the evaluation registry only once its column semantics, incumbent prediction, measurement model, provenance and mechanism-level grouping have been audited.

5 Data availability

The public COBE-FIRAS, Solar-System, Bertozzi, NIST, Millikan and Kamerlingh Onnes inputs are identified in the Supplementary Information. The data package contains source URLs and hashes; the 55-family formula registry and 440 generated point clouds; the 11-family development registry and 88 generated records; the pre-1900 training, calibration and evaluation manifests; and every numerical result used in the figures. The project repository is available at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>, and the archive for this release at <https://doi.org/10.5281/zenodo.22742368>; <https://doi.org/10.5281/zenodo.22138620> resolves to the most recent version.

6 Code availability

Code is available at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>. The repository contains corpus audits, point-cloud generators, frozen evaluators, statistical calibration and robustness experiments, model definitions, training configurations, figure scripts and the evidence-manifest validator. Checkpoint and data hashes are verified before evaluation, and the release archived

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at <https://doi.org/10.5281/zenodo.22742368> is the version used for every number 481
reported here. 482

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8 Author contributions 487

C.H. conceived the framework, designed and built EPOCH, ran the experiments, and wrote 488
the manuscript. J.C. contributed to the benchmark construction, case-study analysis, and 489
manuscript revision. Both authors reviewed and approved the final manuscript. 490

9 Competing interests 491

The authors declare no competing interests. 492

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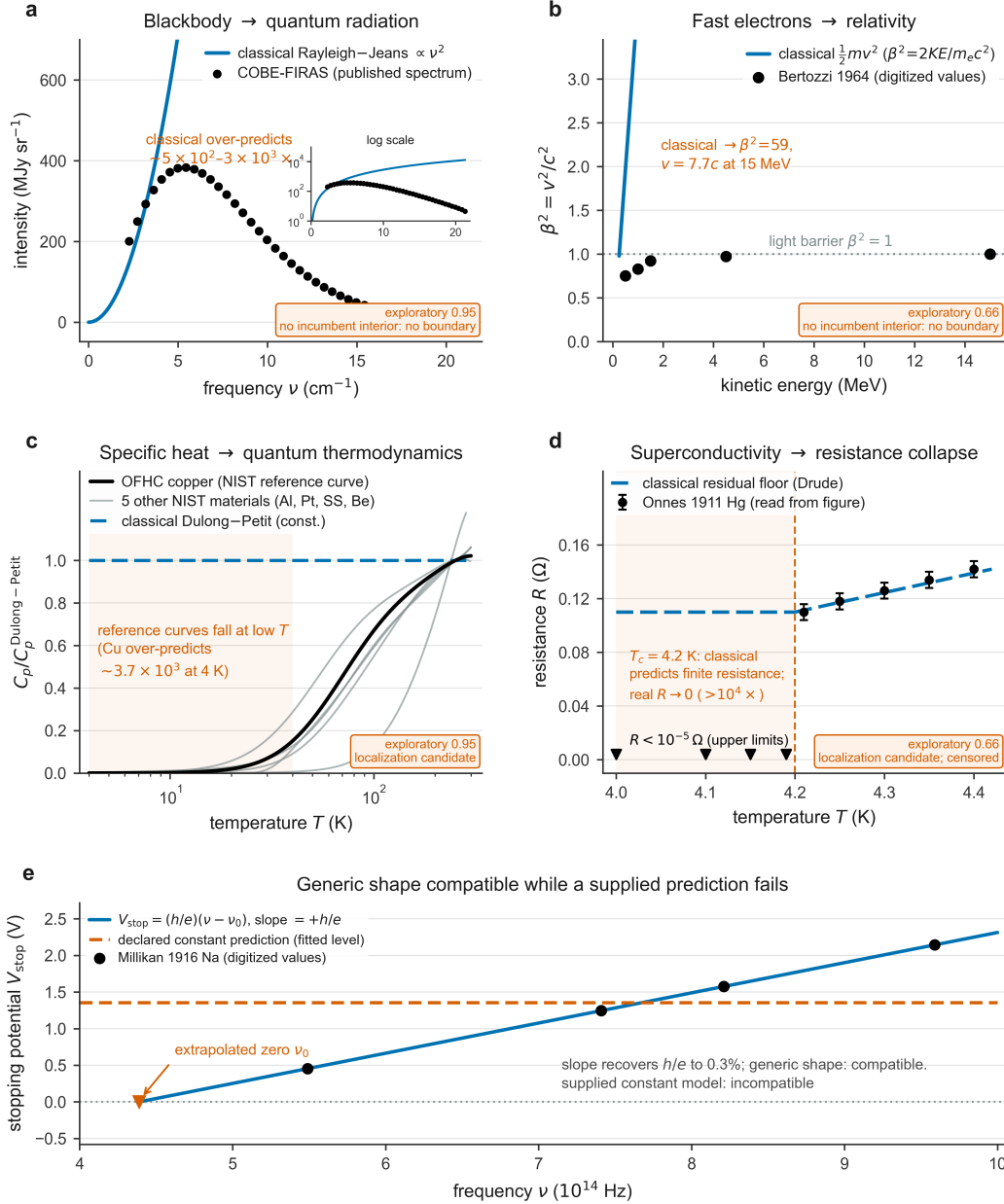


Figure 4. Historical measurements under the 1899 horizon. a–d, Black points show published processed measurements, digitized historical values or evaluated reference curves as labelled; blue curves show incumbent predictions. Orange badges give screening percentiles. FIRAS and Bertozzi yield whole-range model rejection. Specific heat and Onnes contain an incumbent segment, allowing a candidate boundary to be marked by the shaded frontier. Tail probabilities and multiplicity calculations are reported in Supplementary Note 5. e, In Millikan’s 1916 digitized values, stopping potential rises linearly with frequency as $V_{stop} = (h/e)(\nu - \nu_0)$. The slope recovers h/e to 0.3%, and the marker denotes the extrapolated zero crossing ν_0 . The generic shape score clears the line, whereas the theory-conditioned component rejects the classical prediction of frequency-independent energy (dashed line).

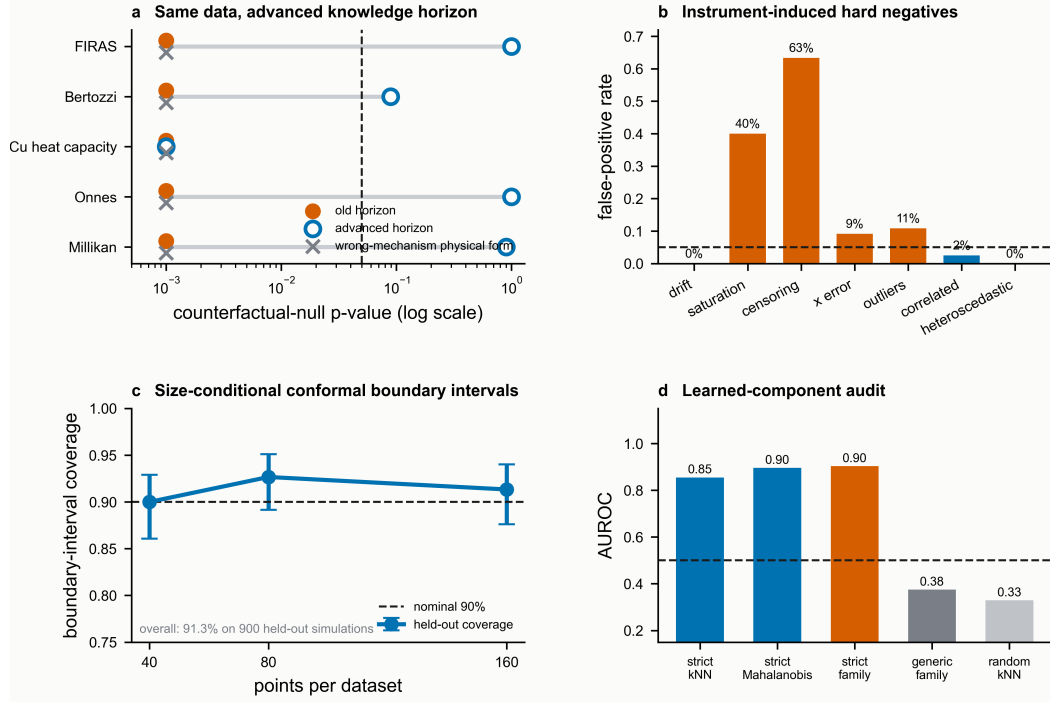


Figure 5. Calibration and controls for the EPOCH components. **a**, Same-grid goodness-of-fit p -values under old and advanced horizons. Four of five advanced predictions clear at 5%; five wrong-mechanism forms remain rejected. **b**, False-positive rates under simulated instrument effects, with the largest changes caused by saturation and censoring. **c**, Held-out coverage of size-conditional 90% split-conformal boundary intervals. **d**, Positive-only learned audit after source reconstruction. The zero-unit encoder reaches synthetic AUROC 0.854 by kNN and 0.895 by shrinkage Mahalanobis; family-grouped AUROC is 0.903 [0.729, 1.000], against 0.329 and 0.288 for matched random features. Training uses 205 records from 40 pre-1900 families and no breakdown examples. At the frozen family threshold it detects 2/12 breakdown families while clearing all classical controls. The matched generic-function model reaches family AUROC 0.375 [0.167, 0.618] on the same evaluation families, a paired difference of 0.528 [0.264, 0.764]; across three paired seeds pre-1900 and generic AUROCs range over 0.889–0.910 and 0.292–0.403.

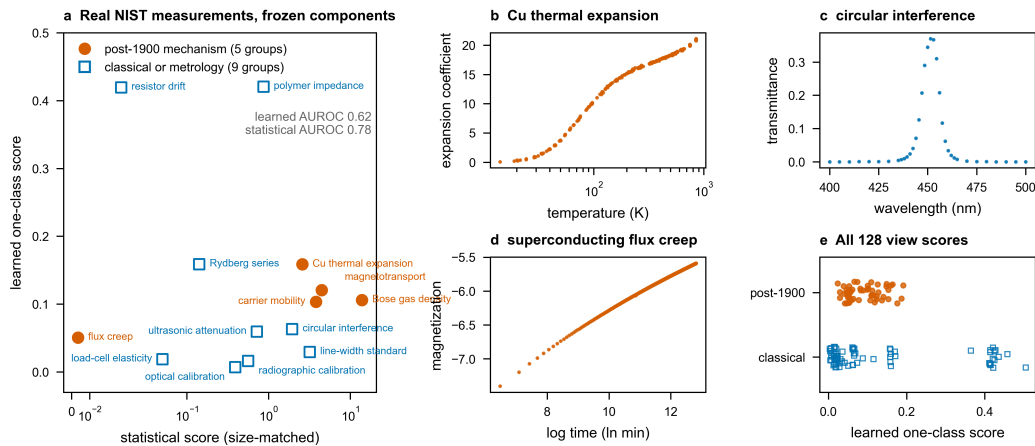


Figure 6. Frozen components on measured series. **a**, Mechanism-group medians of the learned and four-form scores for sixteen NIST measurement series in fourteen groups, five outside the pre-1900 model set and nine classical or metrological. **b–d**, Three of the measured series: copper thermal expansion from 24 K to 852 K, circular-interference transmittance, and superconducting magnetization relaxation against log time. **e**, Every individual measurement-view score behind the medians in **a**. The released quantity is the continuous component output.

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